



TAS LAB

Trustworthy AI and
Autonomous Systems Lab

THE HONG KONG
POLYTECHNIC UNIVERSITY
香港理工大学

Department of
Aeronautical and Aviation Engineering
航空及民航工程學系

Remarkable Team Milestones in 2023

- Dr Weisong Wen appointed Assistant Professor, PolyU AAE (Sep 2023)
- First listing among the World's Top 2% Scientists (Stanford/Elsevier)
- 5 papers at IEEE ITSC 2023 and the 2nd IV Meets Urban workshop

2023 TASLAB Annals

Webpage: <https://polyu-taslab.github.io>

Trustworthy AI and Autonomous Systems Lab, PolyU

3

journal papers

17

conference papers

10

members & visitors

580

Google Scholar citations

4

awards & recognitions

54

news stories on our website

Faculty & Staff



Dr Weisong Wen
Research Asst Prof → Asst
Prof, AAE (Sep 2023)

PhD/MPhil Student Members (and Undergraduates)



Xi ZHENG
PhD



Xikun LIU
PhD



Yihan ZHONG
PhD



Runzhi HU
PhD



Feng HUANG
PhD



Peiwen YANG
PhD (2023â€")



Yifeng CAO
PhD (2023â€")



Ruijie XU
PhD (2023â€")



Yuanyuan ZHANG
PhD (2023â€")

2023 in Review

2023 laid the foundations of the lab. In September Dr Weisong Wen moved from Research Assistant Professor to Assistant Professor at PolyU AAE, and he was listed for the first time among the World's Top 2% Scientists. The team published 3 journal and 17 conference papers, presented five papers at IEEE ITSC 2023 in Bilbao and ran the second "Intelligent Vehicle Meets Urban" workshop there. New collaborations started with Leibniz Universität Hannover, Tencent and PolyU's deep-space and data-science research centres, and the GLIO GNSS/LiDAR/IMU system was accepted by IEEE T-IV and released as open source.

Selected papers accepted in 2023

1. X. Liu, W. Wen, L.-T. Hsu. GLIO: Tightly-coupled GNSS/LiDAR/IMU integration for continuous and drift-free state estimation. IEEE T-IV
2. W. Wen, X. Bai, L.-T. Hsu. 3D vision aided GNSS real-time kinematic positioning for autonomous systems in urban canyons. NAVIGATION 70(3)
3. L. Zhang, W. Wen, L.-T. Hsu, T. Zhang. An improved inertial preintegration model in factor graph optimization. IEEE T-IV
4. F. Huang, W. Wen, J. Zhang, C. Wang, L.-T. Hsu. Dynamic object-aware LiDAR odometry aided by joint weightings estimation. IEEE T-IV
5. P. Chen, W. Guan, F. Huang, Y. Zhong, W. Wen, L.-T. Hsu, P. Lu. ECMD: An event-centric multisensory driving dataset for SLAM. IEEE T-IV

Research & Innovation Awards

World's Top 2% Scientists

Stanford University / Elsevier · first listing



Workshop Chair, IEEE ITSC 2023

2nd "Intelligent Vehicle Meets Urban" workshop, Bilbao

Session Chair, ION GNSS+ 2023

Institute of Navigation, USA

GLIO accepted by IEEE T-IV

Open-sourced GNSS/LiDAR/IMU integration

International Journal & Conference Publications

ION NAVI NAVIGATION, Journal of the ION (2)	JOURNAL JMSA Journal of Marine Science and Application (1)	CONF. ITSC Conference papers (5)	CONF. ION GNSS+ Conference papers (4)	CONF. ISPRS Conference papers (3)	CONF. PLANS Conference papers (3)	CONF. ION ITM Conference paper (1)	CONF. IPIN Conference paper (1)
--	--	--	---	---	---	--	---

New collaborations & highlights

- Germany/HK Joint Research Scheme with Leibniz Universität Hannover (GNSS–LiDAR)
- New industry collaboration with Tencent on multi-sensor vehicle positioning
- Deep-space robotics collaboration with PolyU ISE (lunar regolith operations)
- Data-driven GNSS RTK/INS project with PolyU COMP (RC-DSAI)
- GLIO open-sourced; 5 papers at IEEE ITSC 2023 on V2X mapping, NLOS correction, UAV MPC and map-based localization

Talks, Visits & Service

Feb CUHK Robotics and AI Seminar · Hong Kong

Mar Nanchang University · Nanchang

Apr ASTRI collaboration meeting · Hong Kong

Apr HEU seminar · Harbin

Jun HONOR visit to PolyU · Hong Kong

Jun University of Macau seminar · Macau

Jul Hosted Dr Guoquan Huang (Delaware) and Dr Chao-Chung Peng (NCKU) · PolyU

Jul Hosted PhD visitors from Leibniz Univ. Hannover and TUMSAT · PolyU

Sep Workshop chair, IEEE ITSC 2023 · Bilbao

Sep Hosted HKUST visitors incl. Dr Jianhao Jiao · PolyU

Oct Huawei Hong Kong · Hong Kong

Nov Hosted GODO UAV team · PolyU

Nov Visited the Yangjiang offshore-wind lab · Yangjiang

Dec Meituan Robotics Annual Meeting · Shenzhen

Dec Tsinghua University guest lecture (online) · Beijing

Invited Talks

Research on 3D LiDAR Aided (3DLA) Urban GNSS Positioning for Robotics Navigation

Dr. Weisong Wen
Research Assistant Professor
Intelligent Positioning and Navigation Laboratory,
Department of Aeronautical and Aviation Engineering,
The Hong Kong Polytechnic University
Talk at CUHK HKCLR/CURI Robotics and AI Seminar Series
13 Feb 2023

CUHK Robotics and AI Seminar (Feb)

Trustworthy Perception and Control for Unmanned Autonomous Systems (UAS): From Unmanned Ground Vehicles to Unmanned Aerial Vehicles

Dr. Weisong Wen
Intelligent Positioning and Navigation Lab (IPNL)
Department of Aeronautical and Aviation Engineering (AAE)
The Hong Kong Polytechnic University
26 Jun 2023, seminar Talk at FST of University of Macau

University of Macau seminar (Jun)

High Accuracy Urban GNSS Positioning for Smart Systems in Urban Canyons (复杂都市环境下的智能系统可信卫星导航)

Dr. Weisong Wen, Xikun Liu, Xiao Xia, Penggao Yan, Xi Zhang, Dr Li-Ta Hsu
Intelligent Positioning and Navigation Lab (智能定位与导航实验室)
Department of Aeronautical and Aviation Engineering (AAE)
The Hong Kong Polytechnic University
27 Oct 2023

Huawei Hong Kong (Oct)

GNSS Positioning for Unmanned Autonomous Systems (UAS) in Complex Urban Canyons

Dr. Weisong Wen
Assistant Professor
Intelligent Positioning and Navigation Lab (IPNL)
Department of Aeronautical and Aviation Engineering (AAE)
The Hong Kong Polytechnic University
29 Dec 2023, Online

Tsinghua University guest lecture (Dec)